Mehmet Yigit Turali

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Research Interests

- Machine Learning (ML)
- Deep Learning (DL)
- Statistical Learning
- Natural Language Processing(NLP)
- Large Language Models (LLMs)
- Computer Vision
- ML for Time Series Forecasting
- Optimization
- Computational Neuroscience
- Computational Biology

Related Coursework

- Neural Networks, Introduction to Machine Learning, Digital Signal Processing, Statistical Learning and Data Analytics, Probability and Statistics, Linear Algebra, Linear Programming, Large Scale Social and Complex Networks: Design and Algorithms, Advanced Neural Networks and Deep Learning, Game Theory, Deep Generative Networks, Reinforcement Learning and Dynamic Programming

Education

University of California, Los Angeles (UCLA) **Sep. 2024 – *present***
M.S./Ph.D. in Electrical and Computer Engineering *Los Angeles, CA*
GPA: 3.94/4.00, **Advisor:** Dr. Vwani P. Roychowdhury

Bilkent University **Sep. 2019 – Jun 2024**
Bachelor of Science in Electrical and Electronics Engineering *Ankara, Turkey*
GPA: 3.83/4.00 *Summa Cum Laude*

Honors & Awards

- **Bilkent University Research Excellence Award:** Since he published papers in distinguished journals throughout his Bilkent life. **Jun. 2024**
- **Bilkent University Responsible Artificial Intelligence Innovation Award:** MathWorks and FIGES have sponsored the 2024 Responsible Artificial Intelligence Competition. The jury, which includes representatives from MathWorks, FIGES and Tubitak Sage, selected winner for the Responsible Artificial Intelligence Innovation Award. **Jun. 2024**
- **Royal Dutch Shell, Shell Cashflow Forecasting Datathon:** Placed 4st in Shell Cashflow Forecasting Datathon among over 1 thousand people. **May. 2023**
- **Bilkent University, High Honour Certificates:** Achieved "High Honor Certificate" since having a GPA of 3.50 or above while carrying at least a minimum course load and having a CGPA of 2.00 or higher. **2020 – 2024**
- **Bilkent University, Comprehensive Undergraduate Scholarship:** Full tuition waiver, a monthly stipend, and accommodation support during B.S. **2019 – 2024**
- **General Directorate of Higher Education Credit and Hostels Institution Top 100 Scholarship:** A monthly stipend, and accommodation support during B.S. **2019 – 2024**
- **Measuring, Selection and Placement Center of Turkey University Entrance Exam Results:** Placed 71st in the National University Entrance Exam among over 2.5 million students **Jul. 2019**
- **Atilim University High School Physics Competition:** Placed 1st in Atilim University 7th High School Physics Competition among 100 student groups. **May. 2019**

Publications 🎓

Under Review

1. **M. Y. Turali***, S. Rajesh*, M. E. Lorasdagi, C. Duan, K. Sharma, and V. Roychowdhury, "PANINI-DR: Controlled Deep Research Through Generative Semantic Workspaces," *Submitted to the 15th International Conference on Learning Representations (ICLR 2027)*, 2026, arXiv:2610.15208.
*Equal contribution.

Conference & Workshop Papers

4. S. Rajesh*, K. Sharma*, T. Monsoor, **M. Y. Turali**, R. Idro, J. Kayaga, R. Sebunya, T. T. Namata, J. N. Pasqua, V. Roychowdhury, and R. Mazumder, "Teaching LLMs to Recommend and Defer in Underrepresented Epilepsy Care," *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
*Equal contribution.

3. S. Rajesh*, P. S. Holur*, **M. Y. Turali**, C. Duan, and V. Roychowdhury, "Panini: Continual Learning in Token Space via Structured Memory," *International Conference on Machine Learning (ICML)*, 2026 (Poster), arXiv:2602.15156.
*Equal contribution.

2. C. M. McCrimmon*, P. Sinha*, Q. Cao*, T. Monsoor*, K. Sharma, **M. Y. Turali**, R. A. Samarasinghe, and V. Roychowdhury, "Time-Resolved Neuronal Network Dynamics Distinguish Pathological States in Organoid Models," *2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, DOI: 10.1109/ICASSP55912.2026.11461496.
*Equal contribution.

1. Z. Fei*, **M. Y. Turali***, S. Rajesh*, X. Dai, H. Pham, P. S. Holur, Y. Zhu, L. J. Mooney, Y.-I. Hser, and V. Roychowdhury, "Demo: Customizing Open-Source LLMs for Quantitative Medication Attribute Extraction across Heterogeneous EHR Systems," *NeurIPS 2025 Workshop on GenAI for Health: Potential, Trust, and Policy Compliance*, 2025 (Poster).
*Equal contribution.

Journal Papers

5. M. E. Lorasdagi, **M. Y. Turali**, and S. S. Kozat, "Binary Feature Mask Optimization for Feature Selection," *Neural Computing and Applications*, vol. 37, no. 6, pp. 5155–5167, 2025, DOI: 10.1007/s00521-024-10913-9.

4. S. Ozturk, **M. Y. Turali**, and T. Cukur, "HydraViT: Adaptive Multi-Branch Transformer for Multi-Label Disease Classification from Chest X-Ray Images," *Biomedical Signal Processing and Control*, vol. 100, 106959, 2025, DOI: 10.1016/j.bspc.2024.106959.

3. **M. Y. Turali**, A. T. Koc, and S. S. Kozat, "Optimal Stochastic Gradient Descent Algorithm for Filtering," *Digital Signal Processing*, vol. 155, 104731, 2024, DOI: 10.1016/j.dsp.2024.104731.

2. **M. Y. Turali**, M. E. Lorasdagi, and S. S. Kozat, "AFS-BM: Enhancing Model Performance through Adaptive Feature Selection with Binary Masking," *Signal, Image and Video Processing*, vol. 18, no. 11, pp. 7571–7582, 2024, DOI: 10.1007/s11760-024-03411-x.

1. E. Ilhan, **M. Y. Turali**, and S. S. Kozat, "Gradient Boosting With Moving-Average Terms for Nonlinear Sequential Regression," *IEEE Signal Processing Letters*, vol. 30, pp. 1182–1186, 2023, DOI: 10.1109/LSP.2023.3309577.

Preprints

1. S. Turk, A. Demirkaya, **M. Y. Turali**, C. Hepdurgun, S. U. H. Dar, A. K. Karabulut, A. Azizova, M. Orman, I. Tamsel, U. Aydingoz, M. Argin, and T. Cukur, "JointNET: A Deep Model for Predicting Active Sacroiliitis from Sacroiliac Joint Radiography," *arXiv preprint arXiv:2301.10769*, 2023, DOI: 10.48550/arXiv.2301.10769.

Academic Duties

Reviewer – NeurIPS 2026 Workshop on GenAI for Health (GenAI4Health) **Sep. 2026**
The Third Workshop on GenAI for Health: Agentic Systems, Clinical Trust, and Future Potential

Academic Mentor – IPAM RIPS (OpenAI Project Team) **Jun. 2026 – Aug. 2026**
Institute for Pure and Applied Mathematics (IPAM), UCLA

- Served as the Academic Mentor for the **OpenAI team** at the IPAM Research in Industrial Projects for Students (RIPS) program, partnering with OpenAI's scientific mentor to advise top undergraduates on an OpenAI-sponsored summer research project.

Graduate Student Researcher **Sep. 2024 – Present**
University of California, Los Angeles

Teaching Assistant **Jan. 2025 – Present**
University of California, Los Angeles

- ECE C147/C247A – Neural Networks and Deep Learning (Fall 2026)
- ECE M146 – Introduction to Machine Learning (Spring 2026)
- ECE C147 – Neural Networks and Deep Learning (Winter 2026)
- ECE C143 – Neural Signal Processing (Fall 2025)
- ECE 131A – Probability & Statistics (Winter 2025)

Undergraduate Research Assistant **Jun. 2021 – Jun. 2024**
Bilkent University

Undergraduate Teaching Assistant **Feb. 2021 – Jan. 2022**
Bilkent University

- CS 115 - Introduction to Programming in Python
- EEE 102 - Introduction to Digital Circuit Design

Research Experience

Generative Semantic Workspace (GSW) & Deep-Research Agents **June. 2024 – Present**
Graduate Student Researcher, UCLA Electrical and Computer Engineering

- Leading **PANINI-DR** (co-first author; **under review at ICLR 2027**), a multi-agent deep-research harness that separates coordination of the overall investigation from research and verification of individual subquestions, with a shared **GSW workspace** recording subquestions, dependencies, findings, and supporting evidence.
- Designed a dedicated **verification agent** that checks proposed answers against their subquestions and evidence; its feedback steers further investigation, and ablations show that structured planning, persistent state, and evidence-checked verification each improve performance.
- Achieved the strongest average performance on BrowseComp-Plus, FRAMES, and MoNaCo against seven baselines (shared LLM backbone, frozen corpora): **73.3%** accuracy on BrowseComp-Plus (**+16.0 pp** over the next-best baseline), 26 of 50 hard questions solved (nearly twice the next-best), and the highest answer recoverability from saved evidence on all three benchmarks.
- Co-developed **PANINI** (**accepted to ICML 2026, Poster**), a non-parametric continual-learning framework that encodes documents into GSWs—entity- and event-aware networks of atomic question–answer pairs—and retrieves evidence with **Reasoning Inference Chain Retrieval (RICR)**, a beam search over multi-hop inference chains backed by dual BM25 + dense indexing.
- PANINI achieved top average performance across six QA benchmarks with **2–30× fewer** answer-context tokens than competitive baselines and fewer unsupported answers on curated unanswerable queries, without continual parameter updates.

NLP and LLM Applications for Genomic Structural Variant Detection **Sep. 2024 – Present**
Graduate Student Researcher, UCLA Electrical and Computer Engineering

- Working under the supervision of Prof. Vwani P. Roychowdhury on applying Natural Language Processing (NLP) and Large Language Model (LLM) techniques to genomic data analysis.
- Developing novel one-shot structural variant detection methodologies using transformer-based architectures that treat DNA sequences as a specialized form of language.
- Implementing Reference-Free DNA Embedding (RDE) models inspired by NLP techniques like contrastive learning to create semantically meaningful representations of genomic sequences for direct variant identification.

Brain Organoid Computing Research Project **Sep. 2024 – Present**

Graduate Student Researcher, UCLA Electrical and Computer Engineering

- Co-authored an **ICASSP 2026** paper on a time-resolved network analysis pipeline for two-photon calcium imaging of brain assembloids: dynamic network features (degree variance, clustering) separate an Alzheimer's-linked *MAPT* variant from isogenic controls (Random Forest F1 = 0.90).
- Conducted pioneering research on brain organoid computing under the supervision of Prof. Vwani P. Roychowdhury, exploring the computational capabilities of biological neural networks
- Developed and implemented protocols for interfacing brain organoids with microelectrode arrays to harness their inherent computational properties for complex data processing
- Currently focused on applying organoid-based reservoir computing approaches to forecast chaotic time series, achieving promising preliminary results compared to traditional deep learning methods
- Collaborated with an interdisciplinary team to optimize stimulation parameters and recording techniques for enhancing the predictive capabilities of organoid neural networks in nonlinear dynamical systems

GlobalHealthLLM Medical Research Project **Jan. 2025 – Present**

Graduate Student Researcher, UCLA Electrical and Computer Engineering

- Co-developed **MANANA** (**accepted to NeurIPS 2026**), a non-parametric prompt-learning framework (single- and multi-agent variants) that converts observed prescription errors into auditable prompt memories, adapting LLMs to local anti-seizure medication prescribing practice from longitudinal, unstructured clinic notes.
- MANANA's **Bayesian prompt averaging** turns the learned prompt trajectory into prescription likelihoods and an uncertainty-based deferral signal: on an independently collected held-out Ugandan cohort it improves visit-level top-3 accuracy by **4–8 pp** over prompt-optimization baselines and auto-handles the most confident half of cases at **95% precision** (quarter at 99%), deferring the rest for specialist review.
- Leading a multidisciplinary research initiative focused on leveraging large language models (LLMs) to refine and standardize medical records from underserved communities in Uganda
- Developing and implementing machine learning models to predict optimal anti-seizure medication efficacy for pediatric epilepsy patients in Ugandan tribal communities
- Creating an LLM-powered system that transforms unstructured clinical notes into standardized medical records while preserving culturally specific health information and treatment approaches
- Collaborating with local healthcare providers to design and deploy an accessible interface for treatment decision support, aiming to improve medication adherence and seizure control in rural settings

AI Chatbot Assistant Agents Research Project **Jan. 2024 – Jun. 2024**

Machine Learning Researcher, DataBoss Inc.

- Worked under the supervision of Prof. S. Serdar Kozat on the development of AI-powered chatbot and agent assistants for various applications using Langchain.
- Created and developed AI-driven chatbots and agent assistants tailored for different purposes, utilizing Langchain to enhance their capabilities.
- Developed various Retrieval-Augmented Generation (RAG) techniques and agent architectures, including self-multi RAG based on self-RAG, to improve information retrieval and response generation for AI assistants.

Time Series Forecasting by Using ML and DL Research Project **Dec. 2022 – Jan. 2024**

Machine Learning Researcher, DataBoss Inc.

- Worked under the supervision of Prof. S. Serdar Kozat on time series analysis and forecasting with Statistical Learning and Machine Learning Algorithms in the renewable energy and finance sector.
- Developed a novel algorithm that integrates error terms into Gradient Boosting Machines as a feature for forecasting renewable energy generation and making decisions in the renewable energy sector.

Unmanned Aerial Vehicle Research Project **Jun. 2022 – Aug. 2022**

Embedded Software Research Intern, BAYKAR Inc.

- Took an active role in a project focused on autonomous locating UAVs, serving both BAYKAR and the Turkish Armed Forces.
- Developed an embedding software pipeline for UAVs to reduce the need for human control.

Diagnosing Various Diseases from X-ray Images with DL Research Project **Jun. 2021 – Dec. 2022**
Computer Vision Researcher, National Magnetic Resonance Research Center

- Assisted Prof. Dr. Tolga Çukur and Ph.D. student Ahmet Demirkaya on a master's degree project about diagnosing sacroiliitis (which is an inflammation of one or both sacroiliac joints) from X-ray images by using deep neural network systems. These networks can diagnose the disease from X-ray images that medical doctors would not be able to diagnose without an MRI image.
- Assisted Prof. Dr. Tolga Çukur and Asst. Prof. Dr. Şaban Öztürk on a Tübitak project about diagnosing 14 different chest diseases from X-ray images by using deep neural network systems. These networks can diagnose disease from X-ray images without medical doctors' supervision.

Other Projects

Wind Energy Forecasting

Leveraged SCADA data to forecast wind energy. Applied multiple regression techniques including LASSO, decision trees, and neural networks. Enhanced model accuracy with feature engineering, integrating lag features and a rolling window mean strategy for time series data.

Text-to-Image Synthesis

Conducted text-to-image synthesis using GAN architectures: DCGAN, DALL-E, and DFGAN. While DALL-E generated higher-resolution images, DFGAN faced GPU constraints. DALL-E stood out due to its performance with fewer epochs.

Game "Shoot and Run!" on BASYS3

Developed a game "Shoot and Run!" for the BASYS3 FPGA board using VHDL. Displayed on a VGA monitor, the game features spaceship combat, player movement via pushbuttons, level selection through switches, and player stats on LEDs.

TRC-10 Transceiver

Engineered a TRC-10 transceiver operating at a wavelength of 10 meters.

Technical Skills

- **Programming Languages:**
  - Python, MATLAB, Verilog HDL
- **Machine Learning Libraries:**
  - Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly & Folium, ElasticNet
- **Deep Learning Frameworks:**
  - TensorFlow, PyTorch, Keras, Transformers, Hugging Face, VLLM
- **Vector Databases & Retrieval:**
  - Pinecone, FAISS, Langchain
- **Neuroscience Analysis Tools:**
  - Elephant, Kilosort, SpikeInterface
- **Tools & Version Control:**
  - Git & GitHub, Docker & LaTeX